

From Simple to Multiple Regression in Global Health

Confounding, Causality, and AI Perspectives (One Health)

Pinto Jonathan Katende

2026-05-05

Table of contents

1. Introduction	2
1.1 Research Question	2
2. Data Simulation: A One-Health Perspective	2
3. The Naive Approach	3
3.1 Simple Linear Regression	3
3.2 Visualization	4
4. Hidden Structure: Confounding	5
4.1 Correlation Between Predictors	5
4.2 Interpretation	7
5. The Adjustment	7
5.1 Multiple Linear Regression	7
5.2 Key Insight	8
6. Model Refinement	8
6.1 Variable Selection (AIC)	8
AI Perspective	9
7. Model Diagnostics	10
7.1 Global Diagnostic (Recommended)	10
7.2 Statistical Tests	12
8. Interaction Effects	13
8.1 Motivation	13
8.2 Simulation with Interaction	13
8.3 Interaction Model	13
8.4 Visualization	14
9. Marginal Effects (AI-friendly interpretation)	15
10. Conclusion	16

1. Introduction

1.1 Research Question

Does increasing environmental exposure (e.g., air pollution) increase disease risk?

We begin with a simple model, then progressively uncover hidden biases and complexity.

2. Data Simulation: A One-Health Perspective

```
library(knitr)
library(kableExtra)
library(tidyverse)
library(gtsummary)
library(car)
library(lmtest)
library(ggeffects)
library(ggpubr)
library(performance)
library(see)
library(corrplot)

generate_complex_data <- function(n = 200, seed = 42){
  set.seed(seed)

  age <- round(runif(n, 20, 80))
  BMI <- round(rnorm(n, 25, 4), 1)
  comorbidity <- sample(c("Low", "High"), n, replace = TRUE)
  comorb_numeric <- ifelse(comorbidity == "High", 1, 0)

  # One Health exposure (PM2.5 air pollution)
  air_pollution_level <- round(5 + (0.15 * age) + (0.4 * BMI) +
                                (4 * comorb_numeric) + rnorm(n, 0, 3), 1)

  # Health outcome
  disease_risk <- round(10 + (0.5 * air_pollution_level) +
```

```

      (-0.2 * age) +
      (-5 * comorb_numeric) +
      rnorm(n, 0, 2), 1)

  data.frame(age, BMI, comorbidity, air_pollution_level, disease_risk)
}

data <- generate_complex_data()

kable(head(data), caption = "Table 1: Baseline characteristics of the simulated cohort") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"))

```

Table 1: Table 1: Baseline characteristics of the simulated cohort

age	BMI	comorbidity	air_pollution_level	disease_risk
75	29.8	High	36.2	7.6
76	29.2	High	29.5	5.4
37	21.0	High	23.1	11.1
70	32.4	High	32.6	9.0
59	22.3	High	25.0	4.4
51	25.4	Low	19.8	12.8

3. The Naive Approach

3.1 Simple Linear Regression

```

model_simple <- lm(disease_risk ~ air_pollution_level, data = data)
summary(model_simple)

```

Call:

```
lm(formula = disease_risk ~ air_pollution_level, data = data)
```

Residuals:

```

      Min       1Q   Median       3Q      Max
-9.705 -2.300  0.094  2.846  7.623

```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.23641	1.35430	9.035	<2e-16 ***
air_pollution_level	-0.12527	0.05426	-2.309	0.022 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

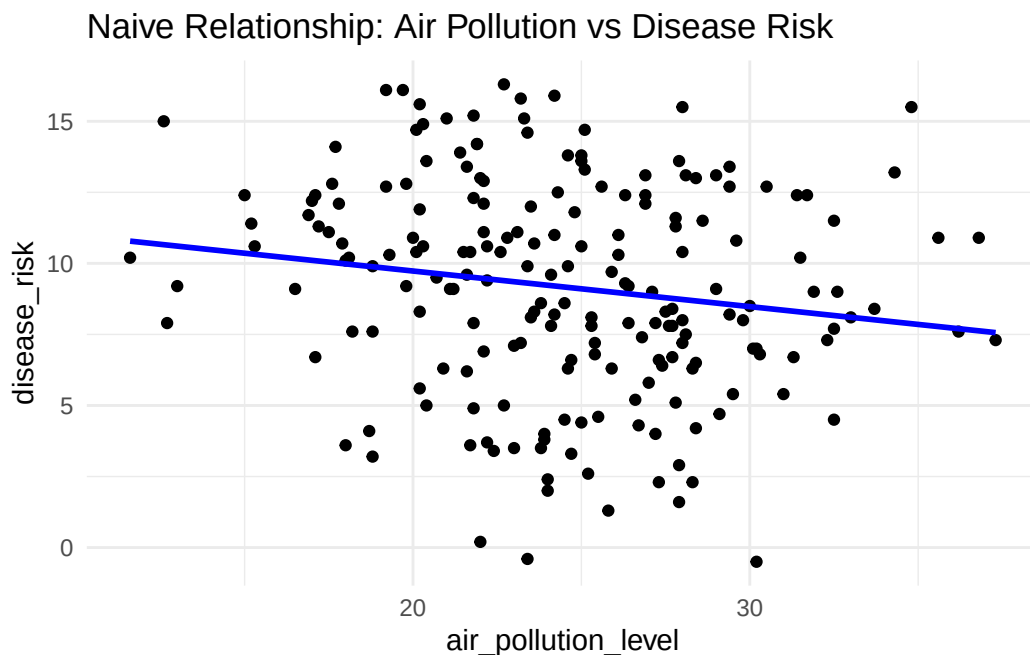
Residual standard error: 3.707 on 198 degrees of freedom

Multiple R-squared: 0.02621, Adjusted R-squared: 0.02129

F-statistic: 5.33 on 1 and 198 DF, p-value: 0.022

3.2 Visualization

```
ggplot(data, aes(air_pollution_level, disease_risk)) +  
  geom_point() +  
  geom_smooth(method = "lm", se = FALSE, color = "blue") +  
  labs(title = "Naive Relationship: Air Pollution vs Disease Risk") +  
  theme_minimal()
```



4. Hidden Structure: Confounding

4.1 Correlation Between Predictors

```
cor(data$air_pollution_level, data$age)
```

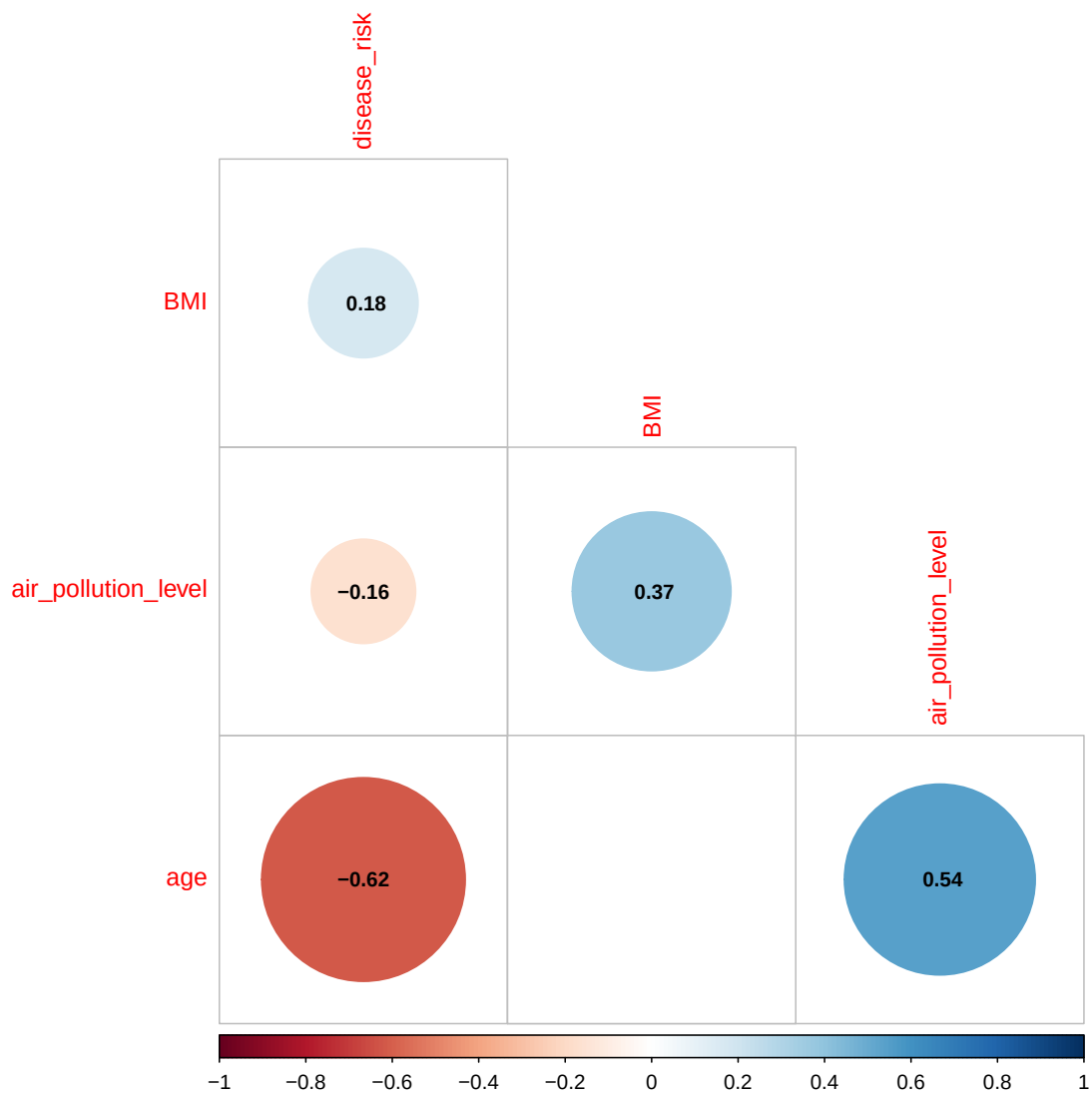
```
[1] 0.5416252
```

```
cor(data$age, data$disease_risk)
```

```
[1] -0.616242
```

```
data_num <- data %>% select(-comorbidity)
cor_matrix <- cor(data_num)
testRes <- psych::corr.test(data_num, adjust = "none")

corrplot(cor_matrix,
          p.mat = testRes$p,
          method = 'circle',
          type = 'lower',
          insig = 'blank',
          addCoef.col = 'black',
          number.cex = 0.8,
          order = 'AOE',
          diag = FALSE)
```



Age is moderately correlated with air pollution exposure ($r = 0.54$) and strongly associated with disease risk ($r = -0.62$), indicating a substantial confounding effect. Adjusting for age is therefore essential to obtain an unbiased estimate of the exposure–outcome relationship

Characteristic	Naive Model			Adjusted Model		
	Beta	95% CI	p-value	Beta	95% CI	p-value
air_pollution_level	-0.13	-0.23, -0.02	0.022	0.50	0.41, 0.58	<0.001
age				-0.20	-0.22, -0.19	<0.001
BMI				0.03	-0.05, 0.11	0.5
comorbidity						
High				—	—	
Low				5.0	4.4, 5.6	<0.001

Abbreviation: CI = Confidence Interval

4.2 Interpretation

Age influences both:

- exposure (air pollution)
- outcome (disease risk)

This creates **confounding bias**

5. The Adjustment

5.1 Multiple Linear Regression

```
model_full <- lm(disease_risk ~ air_pollution_level + age + BMI + comorbidity, data = data)

tbl_merge(
  list(
    tbl_regression(model_simple),
    tbl_regression(model_full)
  ),
  tab_spanner = c("**Naive Model**", "**Adjusted Model**")
)
```

5.2 Key Insight

If the coefficient of air pollution changes → **confounding exists**

6. Model Refinement

6.1 Variable Selection (AIC)

```
model_full <- lm(disease_risk ~ air_pollution_level + age + BMI + comorbidity, data = data)
# Backward selection
model_final <- step(model_full, direction = "backward")
```

Start: AIC=255.14

disease_risk ~ air_pollution_level + age + BMI + comorbidity

	Df	Sum of Sq	RSS	AIC
- BMI	1	1.95	683.27	253.71
<none>			681.32	255.14
- air_pollution_level	1	478.10	1159.42	359.47
- comorbidity	1	907.44	1588.76	422.48
- age	1	1514.28	2195.60	487.18

Step: AIC=253.71

disease_risk ~ air_pollution_level + age + comorbidity

	Df	Sum of Sq	RSS	AIC
<none>			683.27	253.71
- air_pollution_level	1	628.15	1311.42	382.11
- comorbidity	1	933.30	1616.57	423.95
- age	1	1677.22	2360.48	499.66

```
summary(model_final)
```

Call:

```
lm(formula = disease_risk ~ air_pollution_level + age + comorbidity,
```

```

data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-5.6332 -1.2103  0.0716  1.4062  4.8918

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.806336    0.837990   5.736 3.64e-08 ***
air_pollution_level 0.510063    0.037998  13.423 < 2e-16 ***
age           -0.206535    0.009416 -21.934 < 2e-16 ***
comorbidityLow    5.050967    0.308696  16.362 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.867 on 196 degrees of freedom
Multiple R-squared:  0.7555,    Adjusted R-squared:  0.7518
F-statistic: 201.9 on 3 and 196 DF,  p-value: < 2.2e-16

```

AI Perspective

Stepwise selection is a **primitive machine learning approach**.

Modern AI methods use:

- **Lasso (L1 regularization)**
- **Ridge (L2 regularization)**

These methods:

- handle high-dimensional data
- reduce overfitting
- perform automatic feature selection

While `step()` chooses which variables to keep, the Lasso “overwrites” unnecessary variables to zero, which is much more efficient when you have 200 predictors instead of 4.

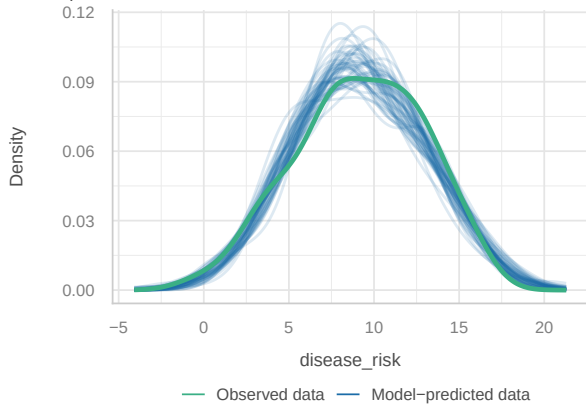
7. Model Diagnostics

7.1 Global Diagnostic (Recommended)

```
check_model(model_final)
```

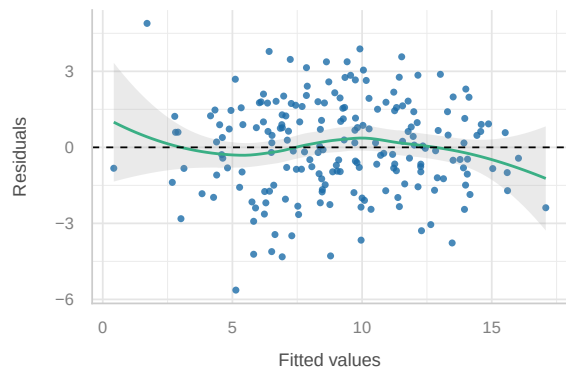
Posterior Predictive Check

Model-predicted lines should resemble observed data line



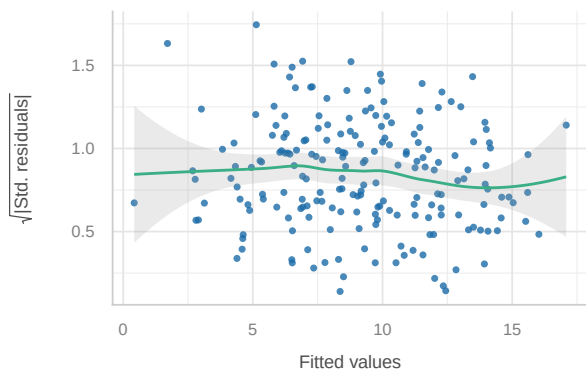
Linearity

Reference line should be flat and horizontal



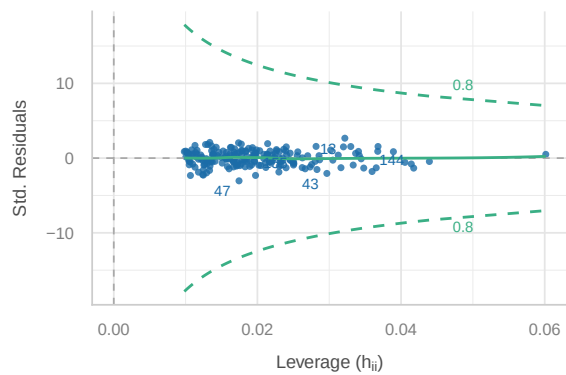
Homogeneity of Variance

Reference line should be flat and horizontal



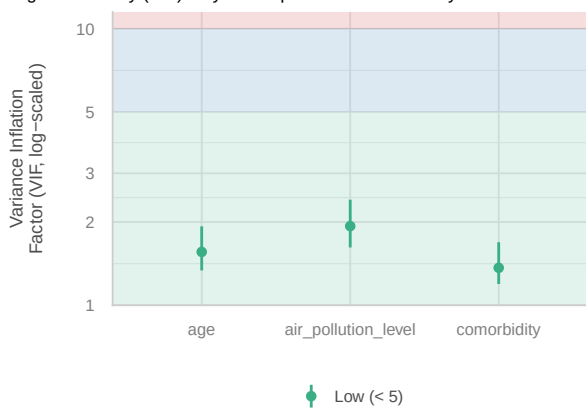
Influential Observations

Points should be inside the contour lines



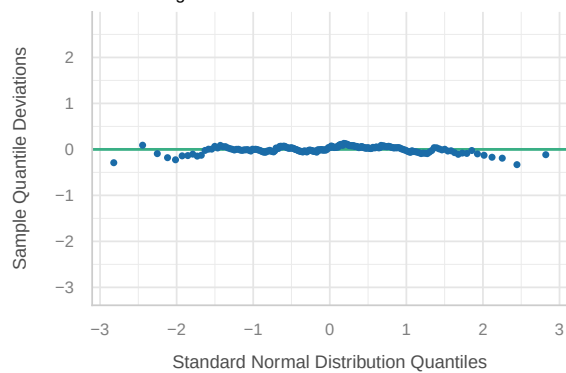
Collinearity

High collinearity (VIF) may inflate parameter uncertainty



Normality of Residuals

Dots should fall along the line



This replaces multiple manual plots with a **clean, unified diagnostic panel**

7.2 Statistical Tests

```
#independence of residuals  
dwtest(model_final)
```

Durbin-Watson test

```
data: model_final  
DW = 1.8924, p-value = 0.2224  
alternative hypothesis: true autocorrelation is greater than 0
```

```
# Homoscedasticity  
bptest(model_final)
```

studentized Breusch-Pagan test

```
data: model_final  
BP = 6.5355, df = 3, p-value = 0.08827
```

```
# Normality  
shapiro.test(resid(model_final))
```

Shapiro-Wilk normality test

```
data: resid(model_final)  
W = 0.99503, p-value = 0.7542
```

```
# Multicollinearity  
vif(model_final)
```

air_pollution_level	age	comorbidity
1.933264	1.559256	1.366231

8. Interaction Effects

8.1 Motivation

Does pollution affect everyone equally?

Maybe not - it may depend on age.

8.2 Simulation with Interaction

```
generate_interaction_data <- function(n = 250, seed = 42){  
  set.seed(seed)  
  
  age <- round(runif(n, 20, 80))  
  comorbidity <- sample(c("Low", "High"), n, replace = TRUE)  
  comorb_numeric <- ifelse(comorbidity == "High", 1, 0)  
  
  air_pollution_level <- 5 + (0.3 * age) + (4 * comorb_numeric) + rnorm(n, 0, 3)  
  
  disease_risk <- 10 + (0.7 * air_pollution_level) +  
    (-0.1 * age) +  
    (-0.01 * air_pollution_level * age) +  
    (-5 * comorb_numeric) +  
    rnorm(n, 0, 2)  
  
  data.frame(age, comorbidity, air_pollution_level, disease_risk)  
}  
  
data_inter <- generate_interaction_data()
```

8.3 Interaction Model

Characteristic	Beta	95% CI	p-value
Air Pollution	0.73	0.58, 0.88	<0.001
Age	-0.06	-0.12, 0.00	0.041
comorbidity			
High	—	—	
Low	4.9	4.3, 5.4	<0.001
Air Pollution * Age	-0.01	-0.01, -0.01	<0.001

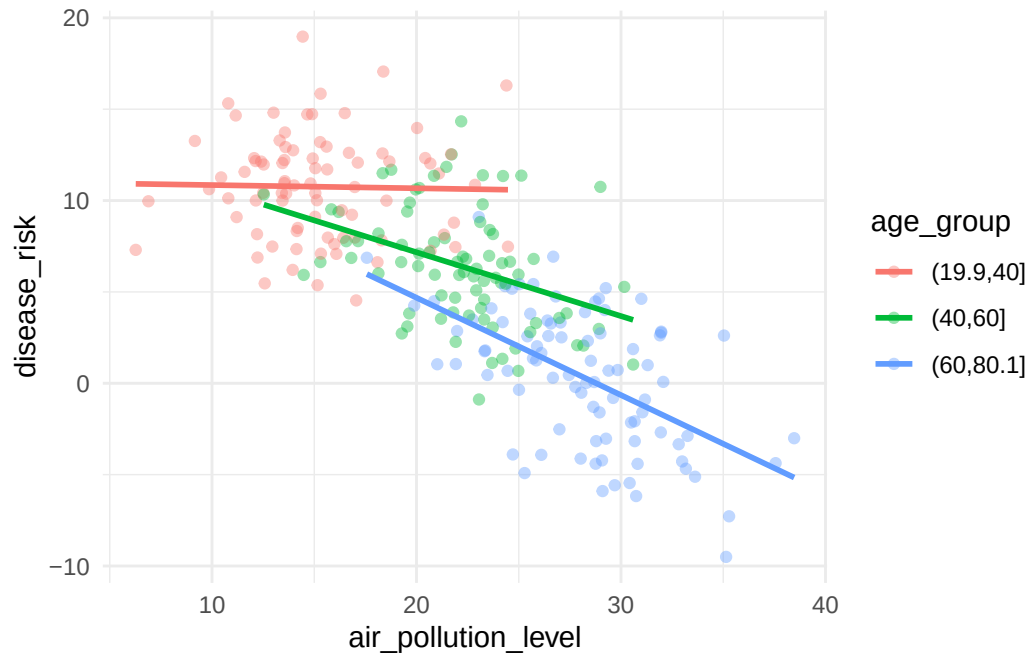
Abbreviation: CI = Confidence Interval

```
model_inter <- lm(disease_risk ~ air_pollution_level * age + comorbidity, data = data_inter)

model_inter %>%
  tbl_regression(
    label = list(
      air_pollution_level ~ "Air Pollution",
      age ~ "Age",
      `air_pollution_level:age` ~ "Pollution × Age Interaction"
    )
  ) %>%
  bold_p()
```

8.4 Visualization

```
data_inter %>%
  mutate(age_group = cut(age, breaks = 3)) %>%
  ggplot(aes(x = air_pollution_level, y = disease_risk, color = age_group)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE) +
  theme_minimal()
```

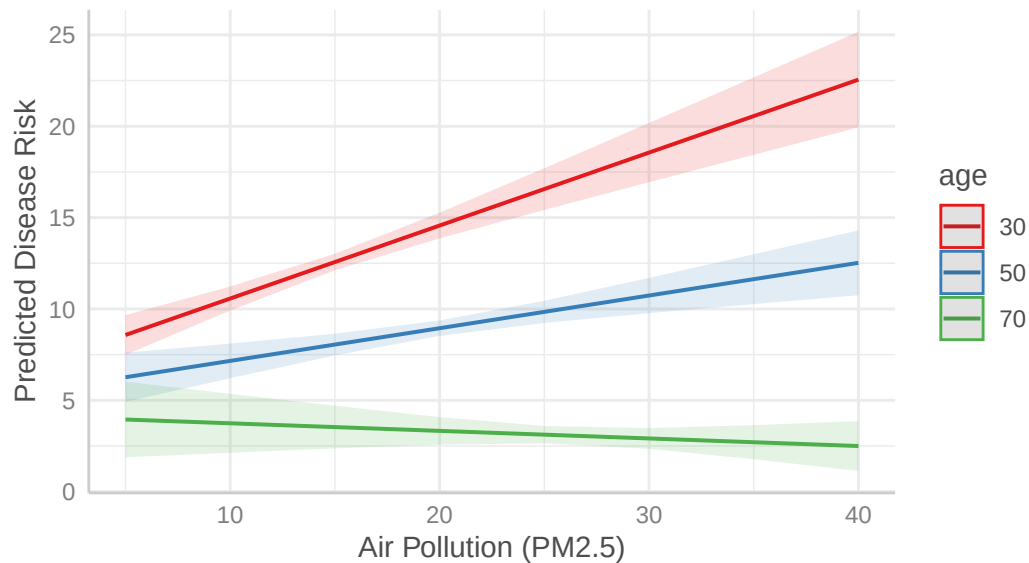


9. Marginal Effects (AI-friendly interpretation)

```
pred <- ggpredict(model_inter, terms = c("air_pollution_level", "age [30,50,70]"))  
  
plot(pred) +  
  labs(  
    title = "Marginal Effect of Air Pollution by Age",  
    subtitle = "Effect decreases with increasing age",  
    x = "Air Pollution (PM2.5)",  
    y = "Predicted Disease Risk"  
  )
```

Marginal Effect of Air Pollution by Age

Effect decreases with increasing age



10. Conclusion

We have shown that:

- Simple models can be misleading
- Confounding biases estimates
- Multiple regression corrects bias
- Interaction reveals population heterogeneity
- AI methods extend these ideas to complex datasets

Final Takeaway

A good model is not the simplest one, but the one that reflects the true data-generating process.